

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 05

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Its total annual discharge	1
01.2	• Public supplies cannot match consumer demand / the water is often cleaner than rivers • It also reduces urban flooding after heavy rain; surplus can be used for aquifer recharge	2
01.3	• The reservoir is created in the estuary so no farmland / settlements need to be flooded • But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
02.1	<i>Any 3 from:</i> • Volume of ash is much less than the original waste (less landfill) • Heat used for district heating / electricity generation (energy recovery) • No sorting / complicated management needed • Destroys pathogens / flammable organic wastes converted to CO ₂ and water • Refuse-derived fuel can replace fossil fuels (eg cement kilns)	3
02.2	• High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) • Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone; also removes heavy metals for storage at high pH • Wet waste needs extra fuel to burn and lowers combustion temperature; composting recovers its nutrient value • Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost	4

Question	Marking guidance	Marks
02.3	- Ash may contain concentrated heavy metals / toxic residues - Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble	2
02.4	<i>Any 1 from:</i> - Fuel (natural gas) needed to burn wet / non-flammable wastes - Cost of treating effluent gases (SO₂, NO_x, HCl, smoke)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
03.1	B - Low liposolubility and low persistence	1
03.2	- Bioaccumulation: build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown - Biomagnification: increase in concentration of the pollutant at successive trophic levels of a food chain	2
03.3	- They are intended to be toxic (to the target species) - They are deliberately released / spread into the environment (rather than being an accidental or waste release)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
04.1	- Ploughing: increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster; carbon in the soil DOM store falls and atmospheric CO₂ rises - Draining: waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly; drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO₂ (though methane release stops) - Marine pollution: kills phytoplankton so less dissolved CO₂ is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton; less CO₂ then dissolves from the atmosphere to replace it, so atmospheric CO₂ rises	6
04.2	<i>Any 3 from:</i> - Rice paddy fields - Landfill sites - Anaerobic sediments in reservoirs - Livestock intestines (cattle) - Manure / slurry stores	3
04.3	More CO₂ dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions; more H⁺ lowers the pH	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.	
05.1	- Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury - Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes - It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water - Local people ate large amounts of fish (critical group) so received toxic doses	4

Question	Marking guidance	Marks
05.2	• Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) • It crosses the placenta and harms the developing unborn baby • Mercury is a teratogen, preventing normal gene expression / development	3
05.3	• Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs • Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut • Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs; liposoluble so most toxic and bioaccumulates	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.	
06.1	• About 8 nights • The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards; extra samples do not change it significantly	2
06.2	• Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size • Sampling at different conditions / times would make results not comparable • Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season)	3
06.3	• Accuracy: how close a recorded result is to the real value • Precision: the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm)	2
06.4	• A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level • Standardised techniques ensure results collected at different times, places or by different people are comparable • Other variables are controlled or monitored so their effect can be assessed; allows the study to be repeated / checked (reliability)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
07.1	• UV light splits diatomic oxygen (O₂) into two monatomic oxygen atoms (O) • Monatomic oxygen reacts with diatomic oxygen to form ozone (O₃) • UV light also splits ozone into O₂ and O; the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat)	3
07.2	• Useful: boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants); dissolve grease but do not damage electrical components (solvents); non-flammable; non-toxic • Persistence: chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere • Dissociation by UV: in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals • Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation; chlorine is regenerated so one atom destroys many ozone molecules; bromine / iodine act similarly	6

Question	Marking guidance	Marks
07.3	• UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) • Damage to leaf tissue and reduced photosynthesis / crop yields (plants) • Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs	3
07.4	• An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution	1
07.5	• Concentration varies with altitude and the stratosphere is not normally sampled; Dobson Units measure the total column thickness instead	1
07.6	• The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
08.1	• A waste is assumed to cause pollution until research shows it is unlikely to do so	1
08.2	• If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution • If polluters can escape their responsibilities / penalties are low, there is little incentive to comply	2
08.3	<i>Any 2 from:</i> • Marine discharges where currents dilute and disperse emissions • Emissions downwind of urban areas • Not discharging waste onto permeable rock above an aquifer	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
09.1	• The climate naturally fluctuates (solar output, Earth's orbit) so extreme events can occur without human influence • An individual event, or even several, does not show a trend; the change may be in the frequency of events which can only be assessed years later • Changes occur at different time and spatial scales (local storm vs global trend); interconnected systems make attribution complex	3
09.2	• Towns have a heat-island effect (heat emissions, dark surfaces absorbing light) • As the town grows around the weather station the warming recorded is partly local, not global; older equipment / sparse global coverage also reduce reliability	2
09.3	• Tree rings: width of each annual ring shows the growth rate, indicating temperature / rainfall in that year; rings easily dated by counting; limitation: growth also affected by other factors (water, pests, shading), only covers the life of the tree / limited regions • Ice cores: annual snow layers compact into ice; air bubbles trapped in the ice give the CO₂ (and methane) concentration when trapped; oxygen isotope ratios indicate temperature; up to 800 000 years (Dome C); limitation: deep layers compressed / less distinct, dating needs radio-isotope analysis and volcanic ash layers; only polar regions • Others accepted: coral growth rings; pollen in lake sediments dated by radiocarbon	5

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 100-102; AQA spec 3.2.1.2.	
10.1	• Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) • They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone • The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface	3
10.2	<i>Any 2 from:</i> • Provides carbon, oxygen, hydrogen and nitrogen (as N₂, O₂, CO₂, H₂O) for biological molecules • Delays the escape of infrared energy, keeping the surface warm • Distributes heat and water vapour by winds / ocean currents • Provides atmospheric pressure so liquid water can exist • Deflects damaging radiation of the solar wind in the upper atmosphere	2
10.3	• Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers	1

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Troposphere: enhanced greenhouse effect - anthropogenic CO₂ (fossil fuels, deforestation), methane (rice, livestock, landfill), N₂O, CFCs, tropospheric ozone; more infrared absorbed and re-emitted; aerosols and smoke increase albedo / scatter light; land-use change alters surface albedo and evapotranspiration • Stratosphere: CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O₂ / O₃ equilibrium; less UV absorbed in the stratosphere, more reaches the surface; stratospheric cooling • Consequences: changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions; increased UV damage to organisms • Difficulties of prediction: linked processes where one change alters others; feedback (positive and negative) and tipping points; time lags (ocean heat capacity, long residence times of CFCs and CO₂); natural variability (solar output, volcanoes, Milankovitch cycles); incomplete understanding of atmospheric processes; limited historical data so proxies needed; models with different assumptions • Evaluation: some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain; precautionary principle • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects • Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) • Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use; life-cycle / embodied energy • Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25